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8. Appendix

Following the closed-world assumption, we use the entities from the PARARULES which includes mainly PEOPLE and ANIMALS. We also consider negation, so there are 4 different scenarios in each depth for each category: PEOPLE with negation in the rules, PEOPLE without negation in the rules, ANIMALS with negation in the rules and ANIMALS without negation in the rules. All of the questions can be derived from the contexts and rules. If a question cannot be matched directly from the context, then it can be derived using rules. For the ANIMALS, we consider 14 different animal entities (“the bald eagle”, “the tiger”, “the bear”, “the lion”, “the wolf”, “the crocodile”, “the dinosaur”, “the snake”, “the leopard”, “the cat”, “the dog”, “the mouse”, “the rabbit”, “the squirrel”), 7 different animals relationships (“is”, “likes”, “chases”, “needs”, “visits”, “attacks”, “sees”), and 28 different animals attributes (“big”, “strong”, “awful”, “fierce”, “heavy”, “horrible”, “powerful”, “angry”, “furry”, “small”, “cute”, “lovely”, “beautiful”, “funny”, “dull”, “rough”, “lazy”, “slow”, “sleepy”, “boring”, “tired”, “reckless”, “kind”, “quiet”, “round”, “nice”, “smart”, “clever”). For the PEOPLE, we consider 9 different entities (“Anne”, “Alan”, “Bob”, “Charlie”, “Dave”, “Erin”, “Harry”, “Gary”, “Fiona”), 1 relationship (“is”) and 20 people attributes (“wealthy”, “smart”, “nice”, “quiet”, “kind”, “poor”, “dull”, “rough”, “bad”, “sad”, “short”, “thin”, “small”, “little”, “big”, “strong”, “high”, “old”, “young”, and “huge”).

Require: Type of dataset: animal/people, add negation rule or not: Yes/No,
reasoning depth $d \in 2, 3, 4, 5$, animal name list = ['the bald eagle', 'the tiger', 'the bear',
'the lion', 'the wolf', 'the crocodile', 'the dinosaur', 'the snake', 'the leopard', 'the cat',
'the dog', 'the mouse', 'the rabbit', 'the squirrel'],
people name list = ['Anne', 'Alan', 'Bob', 'Charlie', 'Dave', 'Erin', 'Harry', 'Gary', 'Fiona'],
animal relation list = ['is', 'is not', 'likes', 'chases', 'needs', 'visits', 'attacks', 'sees'],
people relation list = ['is', 'is not'], animal attribute list = ['kind', 'quiet', 'round', 'nice',
'smart', 'dull', 'rough', 'lazy', 'slow', 'sleepy', 'furry', 'small', 'cute', 'lovely', 'beautiful',
'big', 'strong', 'awful', 'fierce', 'heavy'], people attribute list = ['big', 'strong', 'high',
'huge', 'short', 'thin', 'small', 'little', 'wealthy', 'smart', 'nice', 'quiet', 'kind', 'poor', 'dull',
'rough', 'bad', 'sad', 'old', 'young'] total_list = []
for reasoning depth d in 2,3,4,5 **do**
 item_list = randomly select 4 animals/people name from animal/people name list

 for index in range(0, len(item_list)) **do**
 item, item_1, item_2, item_3 = item_list[0], item_list[1], item_list[2], item_list[3]
 random shuffle animal/people relation list
 if add negation rules == "No" **then**
 context, question, label = load the template and fill item, item_1, item_2, item_3
 to the subject or object in the template, and we select one of the elements from
 the animal/people relation list as the verb. For some facts, we add an item,
 item_1, item_2, item_3 as the subject, and one of the elements from the animal/
 people attribute list as the object to generate a dataset that only includes the
 depth = d , which means all questions have the same number of rules to derive
 the answer.
 else
 context, question, label = load the template, the template is similar to the above,
 but add a negation(not) in any rule to generate a dataset which only includes the
 depth = d
 end if
 total_list = total_list.append(context, question, label, depth = d)
 end for
end for
return total_list

Algorithm 1: PARARULE-Plus data generation